

AGENTIC AI

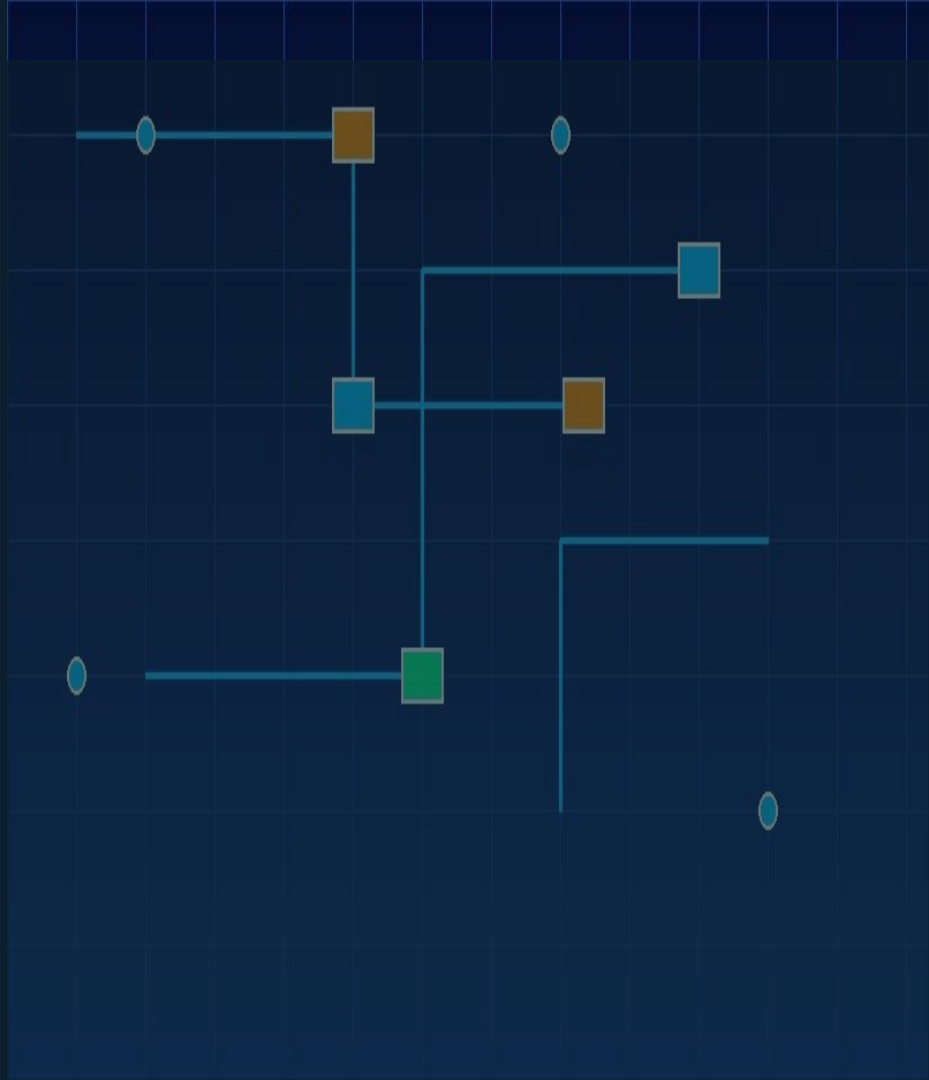
Day 03

Core Components

Model · Tools · Memory

"Today we understand what makes an AI agent intelligent internally."

DAY 03



Learning Objectives



Model Layer

Understand the LLM backbone that powers every AI agent



Tools

How tools are invoked by agents to interact with the real world



Application

Apply these concepts in real-world agent design



Memory

Short-term vs Long-term memory types and strategies

Learning Outcomes

EXPLAIN

Core components of AI agents

IDENTIFY

Tools required in different scenarios

APPLY

Memory strategies in AI systems

EVALUATE

Agent architecture decisions

Ask: "Which component do you think is most important?"

Recap: Day 02

Yesterday We Learned

- Use Cases of Agentic AI
- Types of Agents
- Decision-making flow (Think → Plan → Act)
- Future trends

Today We Discover

WHAT agents do



HOW they work

Model · Tools · Memory

"Yesterday we saw WHAT agents do — today we see HOW they work."

Core Components of AI Agent

These 3 components are the foundation of every AI agent



MODEL

The Brain 🧠



TOOLS

The Hands 🛠️

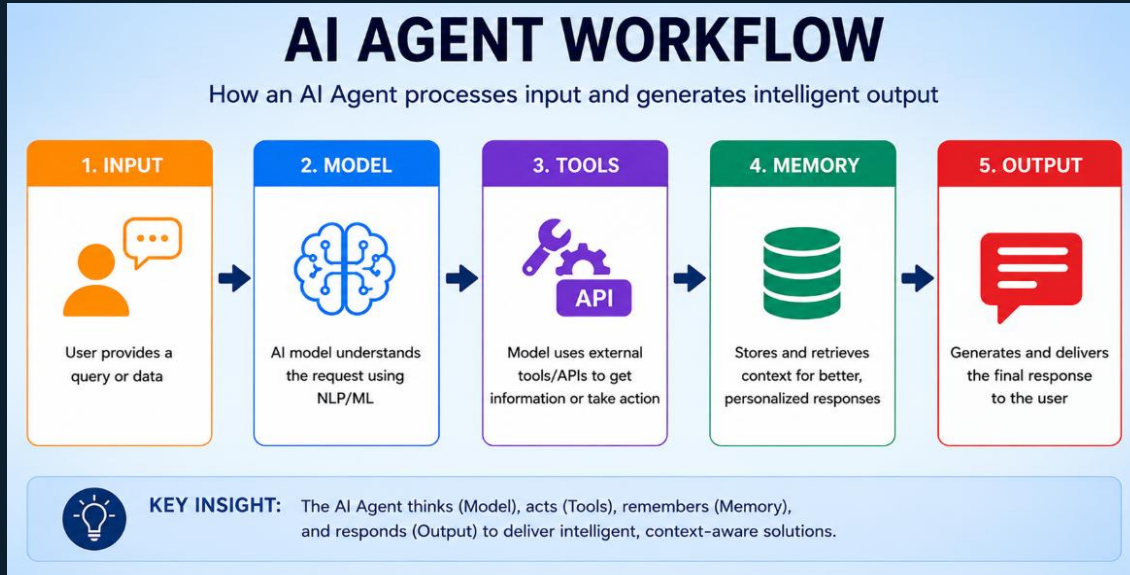


MEMORY

The Experience 📖

"Like a human — Thinking, Acting, and Remembering"

Agent Architecture Overview



Model → Thinking

Tools → Action

Memory → Learning

Every intelligent agent follows this pipeline

Component Breakdown



MODEL

Brain 🧠

Processes input, reasons and generates intelligent output



TOOLS

Hands 🛠️

Executes real-world actions via APIs and external systems



MEMORY

Experience 📄

Stores context and learns from past interactions

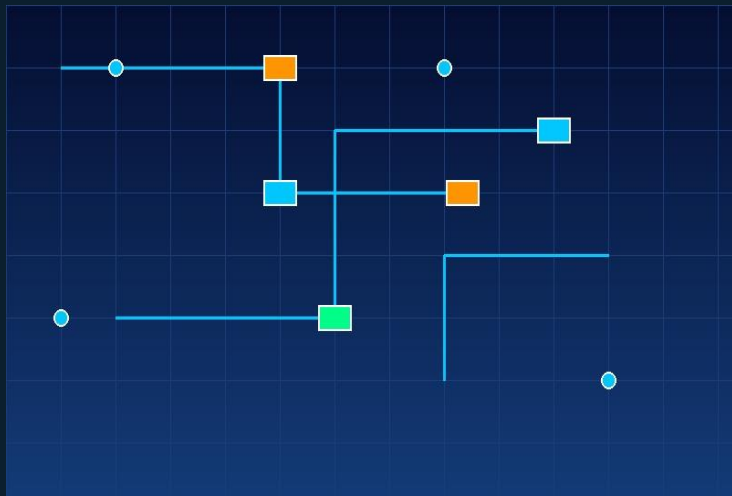
Analogy: Like a human — thinking, acting, and remembering



MODEL — What is it?

What is a Model?

- Core intelligence of the AI agent
- Usually a Large Language Model (LLM)
- Understands and reasons about input
- Generates coherent, intelligent output
- Trained on massive text datasets



It predicts the next best word

based on learned patterns

Model Examples

GPT

by OpenAI

Powers ChatGPT & countless
AI applications worldwide

Gemini

by Google

Google's multimodal flagship
AI model

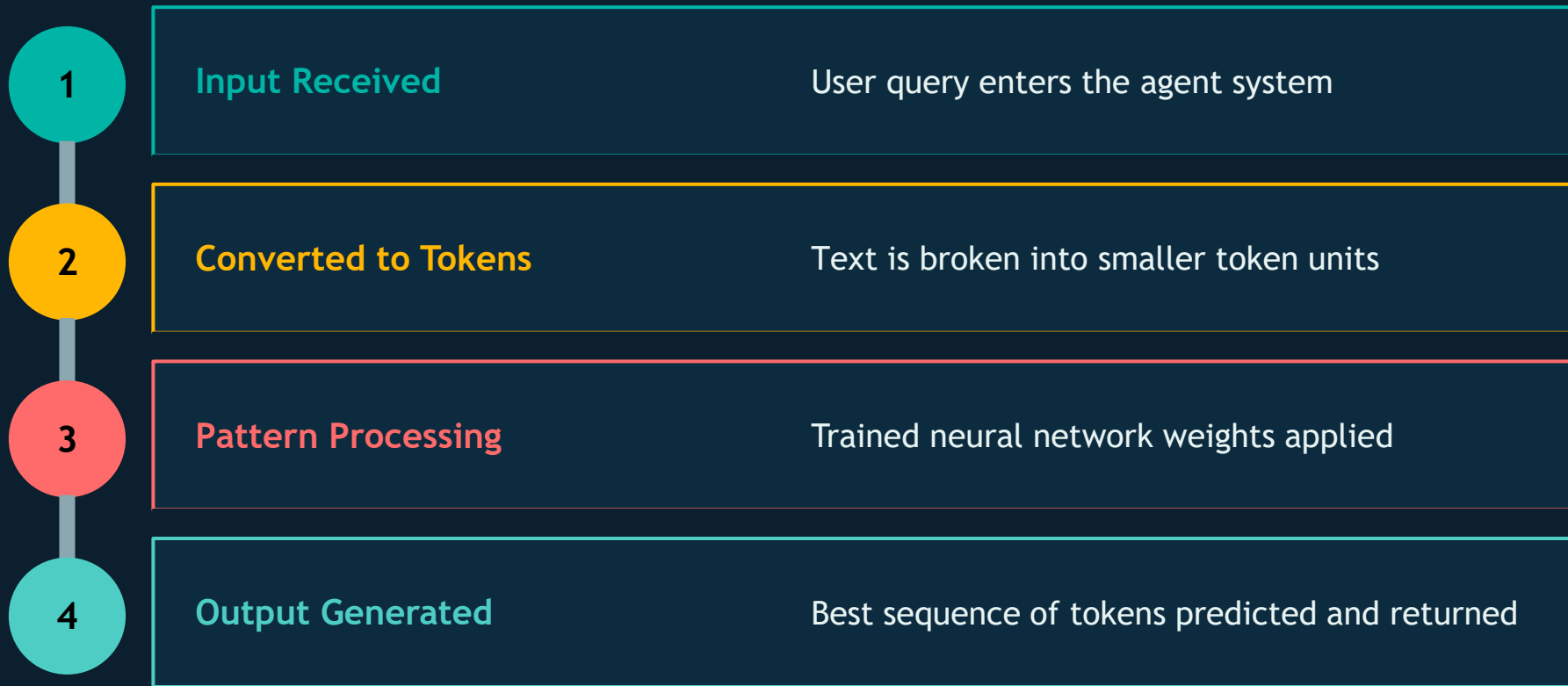
LLaMA

by Meta

Open-source model enabling
community research & apps

These models are trained on billions of text examples

How Model Works



Simple: It predicts the next best word based on learned patterns

Model Capabilities

Answer Questions

Generate Content

Translate Languages

Perform Reasoning

Decision Support



Model Limitations



No real-time data access



Cannot access external systems



Cannot perform real-world actions



Limited memory – context window only

This is why we need Tools + Memory →

Why Model Alone Is Not Enough



Cannot book tickets



Cannot fetch live weather



Cannot remember users



TOOLS

Real-world actions via APIs



MEMORY

Stores user context

QUICK QUIZ

Which task requires more than just a Model?

A

Answer a question

B

Summarize text

C

Book a flight autonomously

D

Translate a sentence

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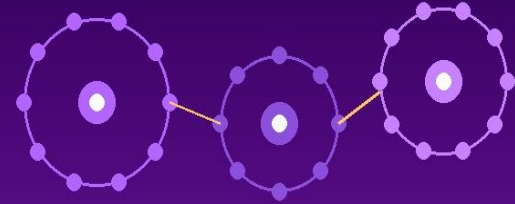
Answer: C — C — Booking a flight requires Tools (APIs) + Model reasoning



TOOLS — What are they?

What are Tools?

- External APIs or callable functions
- Allow AI to interact with real-world systems
- Model decides WHEN to invoke a tool
- Tool returns a result to the model
- Model uses result in its final response



Bridge between AI reasoning

and real-world systems

Tool Examples



Weather API

Fetch live weather data from OpenWeather or NOAA



Payment Gateway

Process transactions via Stripe, Razorpay etc.



Database Query

Read/write structured data from SQL/NoSQL stores

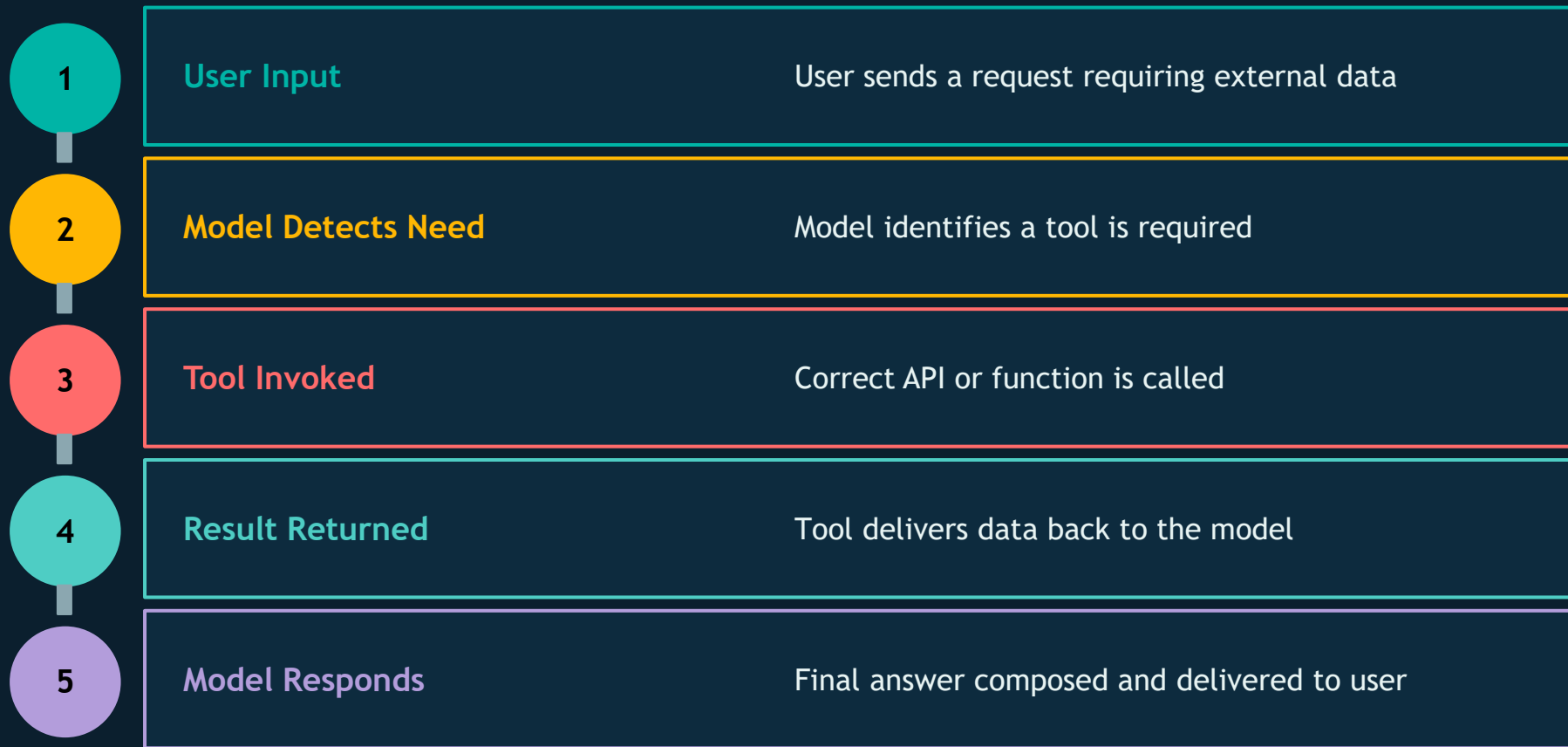


Web Search

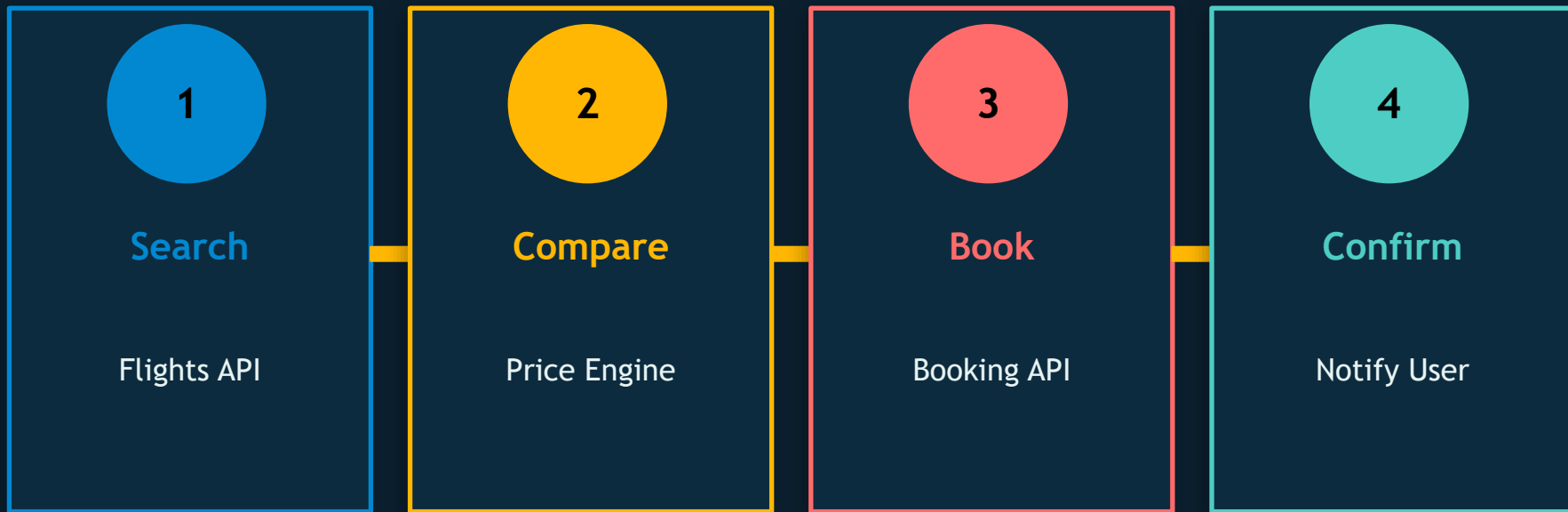
Retrieve real-time information from the internet

Model decides WHEN and HOW to use each tool

Tool Workflow



Real Example: Travel Agent



This complete workflow is IMPOSSIBLE without Tools + Model working together



ACTIVITY

Scenario:

"A chatbot gives live weather updates to users"

What tool is required?



Answer: Weather API



ACTIVITY

Scenario:

"A chatbot gives live weather updates to users"

What tool is required?



MEMORY — What is it?

What is Memory?

- Stores user data and past interactions
- Helps maintain context across sessions
- Enables personalisation and learning
- Two types: Short-Term & Long-Term
- Foundation of agent intelligence



Memory makes agents

personalised & intelligent

Types of Memory

SHORT-TERM

-  Temporary duration
-  Session-based storage
-  Stores current chat history
-  Lost after session ends
-  Example: Active conversation

LONG-TERM

-  Permanent storage
-  Stored in DB / Vector DB
-  Used for personalisation
-  Improves over time
-  Example: User preferences

Memory Comparison

Feature	Short-Term Memory	Long-Term Memory
Duration	Temporary	Permanent
Storage	Session / RAM	Database / Vector DB
Example	Current chat history	User preferences
After session	Lost / cleared	Persisted
Use case	Context in one conversation	Personalisation across sessions

QUICK QUIZ

Which of these is an example of Long-Term Memory?

A

Chat history

B

User preferences

C

Current prompt

D

Session data

QUICK QUIZ

Which of these is an example of Long-Term Memory?

A

Chat history

B

User preferences



C

Current prompt

D

Session data



Answer: B — B — User preferences are stored in databases and persist across sessions

Real-Time Use Case: E-Commerce Agent



Remembers Past Purchases

Long-term memory stores order history across sessions



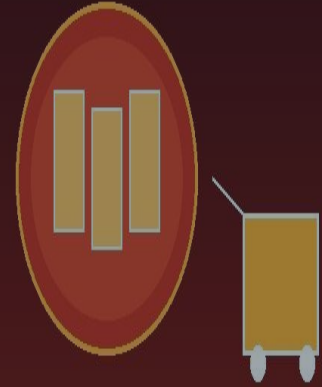
Personalised Suggestions

Model analyses preferences and recommends matching products



Searches Products

Tool calls product catalog & inventory APIs in real-time



Bringing It All Together



MODEL

Think

+



TOOLS

Act

+

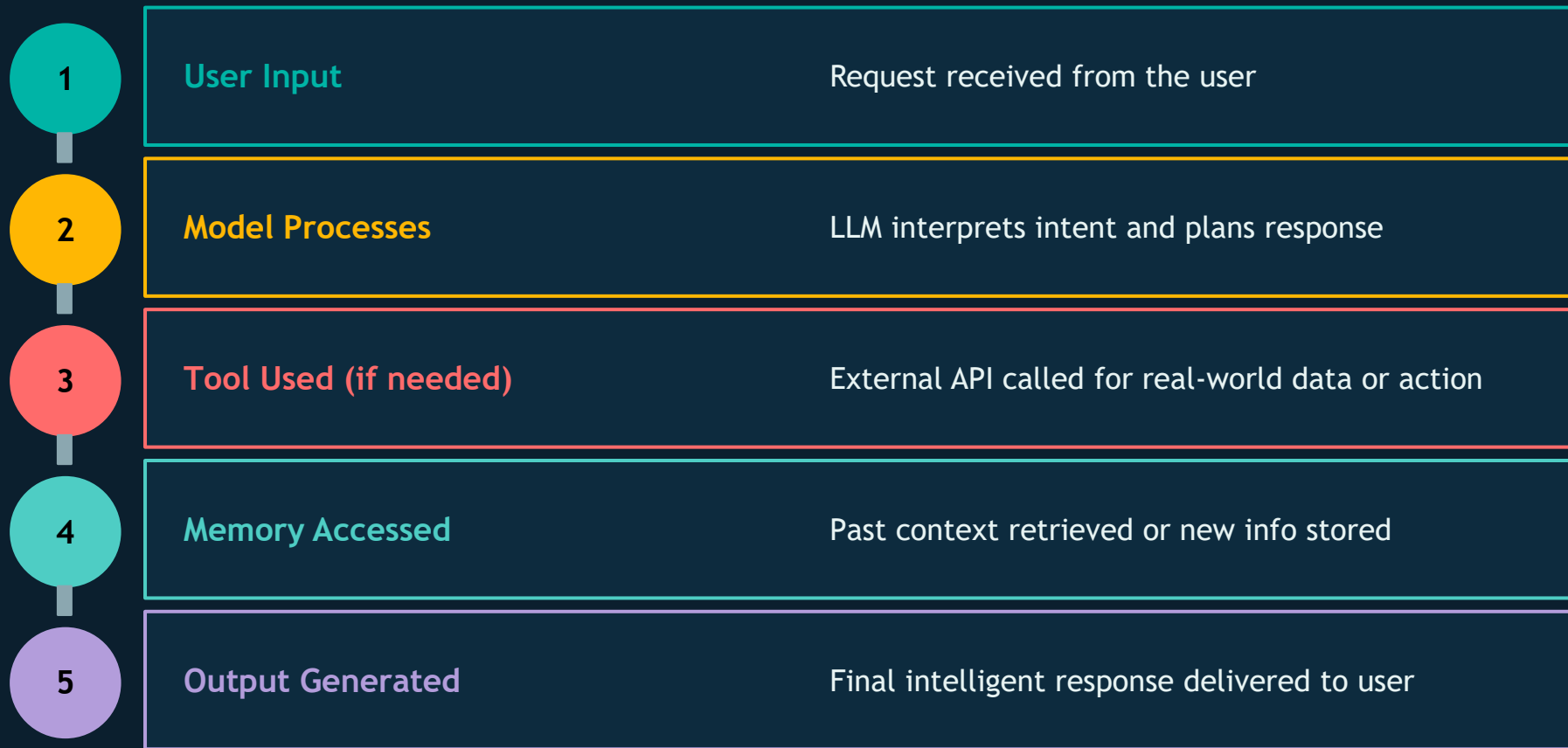


MEMORY

Learn

= Intelligent AI Agent

Full Agent Workflow





FINAL SCENARIO

"Order food based on my past preference"

1



Memory

Retrieves past orders → 'User likes Biryani'

2



Model

Understands intent and decides what to order

3



Tools

Calls food delivery API and places the order

4



Output

"Your Biryani order has been placed!"



Key Insight: Memory → Model → Tools → Output



Scenario Workflow Explained



User Input:

"Order food based on my past preference"



Memory:

Retrieves past orders → "User likes Biryani"



Model:

Understands intent, decides what to order



Tools:

Calls Zomato/Swiggy API, places order



Output:

"Your Biryani order has been placed!"



AGENTIC AI — DAY 03

Today's Takeaways



MODEL = Brain

LLM that understands, reasons & generates output



TOOLS = Hands

APIs & functions that let agents act in the world



MEMORY = Experience

Storage that enables context & personalisation



"These three components are the foundation of every intelligent AI agent"